

Natural_Lacerate

2026-09-27

```
library(tidyverse)
library(janitor)
library(lubridate)
library(grid)
library(emmeans)
library(multcomp)
library(multcompView)
library(car)
library(glmTMB)
library(DHARMA)
library(performance)
library(lme4)
library(patchwork)
library(magick)

rm(list = ls())

project_dir <- "/Users/junbc/Documents/GitHub/heating_lacerates_final/natural"
setwd(project_dir)

select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate
recode <- dplyr::recode

## Figure settings

cohort_levels <- c("W1-2", "W3-4", "W5-6", "W7-8", "W9-10")

cohort_labels <- c(
  "W1-2" = "Week 1-2",
  "W3-4" = "Week 3-4",
  "W5-6" = "Week 5-6",
  "W7-8" = "Week 7-8",
  "W9-10" = "Week 9-10"
)

temp_labels <- c(
  "H2_25" = "25°C",
  "H2_32" = "32°C",
  "H2-25" = "25°C",
  "H2-32" = "32°C"
)
```

```

temp_colors <- c(
  "H2_25" = "#3B6FB6",
  "H2_32" = "#D62728",
  "H2-25" = "#3B6FB6",
  "H2-32" = "#D62728"
)

BASE_SIZE      <- 18
AXIS_TEXT_SIZE <- 14
STRIP_SIZE     <- 16
LEGEND_TITLE   <- 15
LEGEND_TEXT    <- 14
POINT_SIZE     <- 2.8
POINT_ALPHA    <- 0.65
JITTER_WIDTH   <- 0.14
BOX_WIDTH      <- 0.60
BOX_LINEWIDTH  <- 1.20
ANNOT_SIZE     <- 5
PVAL_SIZE      <- 4.5
BRACKET_WIDTH  <- 0.70
TAG_SIZE       <- 18

my_theme <- theme_classic(base_size = BASE_SIZE) +
  theme(
    axis.title = element_text(size = BASE_SIZE),
    axis.text  = element_text(size = AXIS_TEXT_SIZE),
    axis.text.x = element_text(angle = 0, hjust = 0.5, vjust = 0.5),
    strip.text = element_text(face = "bold", size = STRIP_SIZE),
    strip.background = element_rect(
      fill = "white",
      color = "black",
      linewidth = 1.2
    ),
    panel.spacing = unit(1.4, "lines"),
    legend.title = element_text(size = LEGEND_TITLE),
    legend.text  = element_text(size = LEGEND_TEXT),
    plot.tag     = element_text(size = TAG_SIZE, face = "bold")
  )

format_exact_p <- function(p) {
  case_when(
    is.na(p) ~ "p = NA",
    p < 0.0001 ~ "p < 0.0001",
    TRUE ~ paste0("p = ", formatC(p, format = "f", digits = 4))
  )
}

temp_color_scale <- function(values) {
  scale_color_manual(
    values = temp_colors[values],
    labels = temp_labels[values]
  )
}

```

```

temp_x_scale <- function(values) {
  scale_x_discrete(labels = temp_labels[values])
}

cohort_facet <- function(ncol = NULL) {
  facet_wrap(
    ~ cohort,
    nrow = 1,
    ncol = ncol,
    labeller = labeller(cohort = cohort_labels)
  )
}

geom_standard_boxplot <- function(fill = NA) {
  geom_boxplot(
    width = BOX_WIDTH,
    fill = fill,
    linewidth = BOX_LINEWIDTH,
    outlier.shape = NA
  )
}

geom_standard_jitter <- function(height = 0) {
  geom_jitter(
    width = JITTER_WIDTH,
    height = height,
    size = POINT_SIZE,
    alpha = POINT_ALPHA
  )
}

add_p_bracket <- function(data, tick_height) {
  list(
    geom_segment(
      data = data,
      aes(x = 1, xend = 2, y = y_bracket, yend = y_bracket),
      inherit.aes = FALSE,
      color = "black",
      linewidth = BRACKET_WIDTH
    ),
    geom_segment(
      data = data,
      aes(x = 1, xend = 1, y = y_bracket, yend = y_bracket - tick_height),
      inherit.aes = FALSE,
      color = "black",
      linewidth = BRACKET_WIDTH
    ),
    geom_segment(
      data = data,
      aes(x = 2, xend = 2, y = y_bracket, yend = y_bracket - tick_height),
      inherit.aes = FALSE,
      color = "black",
      linewidth = BRACKET_WIDTH
    )
  )
}

```

```

    ),
    geom_text(
      data = data,
      aes(x = 1.5, y = y_text, label = p_label),
      inherit.aes = FALSE,
      color = "black",
      size = PVAL_SIZE
    )
  )
}

## Natural lacerate production

lacerate_meta <- read_csv(
  file.path(project_dir, "data/natural_lacerate_metadata.csv"),
  show_col_types = FALSE
) %>%
  clean_names() %>%
  mutate(
    cohort = factor(cohort, levels = cohort_levels),
    treatment = factor(treatment, levels = c("H2_25", "H2_32")),
    genotype = factor(genotype, levels = "H2"),
    temp = factor(temp, levels = c(25, 32)),
    tub_id = factor(tub_id),
    parent_id = factor(parent_id),
    lacerate_id = as.character(lacerate_id)
  ) %>%
  filter(
    !is.na(treatment),
    !is.na(genotype),
    !is.na(cohort),
    !is.na(lacerate_id),
    lacerate_id != "",
    lacerate_id != "DUPLICATE_REMOVED"
  )

lacerate_unique <- lacerate_meta %>%
  distinct(lacerate_id, cohort, treatment, genotype, temp, tub_id, parent_id)

lacerates_per_tub <- lacerate_unique %>%
  count(cohort, treatment, genotype, temp, tub_id, name = "n_lacerates")

print(lacerates_per_tub)

```

```

## # A tibble: 30 x 6
##   cohort treatment genotype temp tub_id      n_lacerates
##   <fct>   <fct>      <fct>  <fct> <fct>          <int>
## 1 W1-2    H2_25        H2      25   H2_25_Tub1         13
## 2 W1-2    H2_25        H2      25   H2_25_Tub2         12
## 3 W1-2    H2_25        H2      25   H2_25_Tub3          9
## 4 W1-2    H2_32        H2      32   H2_32_Tub1         23
## 5 W1-2    H2_32        H2      32   H2_32_Tub2         18

```

```
## 6 W1-2 H2_32 H2 32 H2_32_Tub3 9
## 7 W3-4 H2_25 H2 25 H2_25_Tub1 33
## 8 W3-4 H2_25 H2 25 H2_25_Tub2 17
## 9 W3-4 H2_25 H2 25 H2_25_Tub3 25
## 10 W3-4 H2_32 H2 32 H2_32_Tub1 28
## # i 20 more rows
```

Statistical letters

```
letters_tub_raw <- lacerates_per_tub %>%
  group_by(cohort) %>%
  group_modify(~ {
    dat <- .x

    if (n_distinct(dat$temp) < 2) {
      dat %>%
        distinct(temp) %>%
        mutate(.group = "a")
    } else {
      mod <- lm(n_lacerates ~ temp, data = dat)
      em <- emmeans(mod, ~ temp)

      multcomp::cld(
        em,
        Letters = letters,
        adjust = "sidak"
      ) %>%
        as.data.frame() %>%
        select(temp, .group) %>%
        mutate(.group = str_trim(.group))
    }
  }) %>%
  ungroup()

positions_tub <- lacerates_per_tub %>%
  group_by(cohort) %>%
  summarise(
    cohort_max = max(n_lacerates, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  crossing(
    temp = factor(c(25, 32), levels = c(25, 32))
  ) %>%
  mutate(y_pos = cohort_max + 5)

letters_tub <- positions_tub %>%
  left_join(
    letters_tub_raw %>% select(cohort, temp, .group),
    by = c("cohort", "temp")
  ) %>%
  mutate(
    treatment = factor(
      paste0("H2_", temp),
      levels = c("H2_25", "H2_32")
    )
  )
```

```

    )
  )

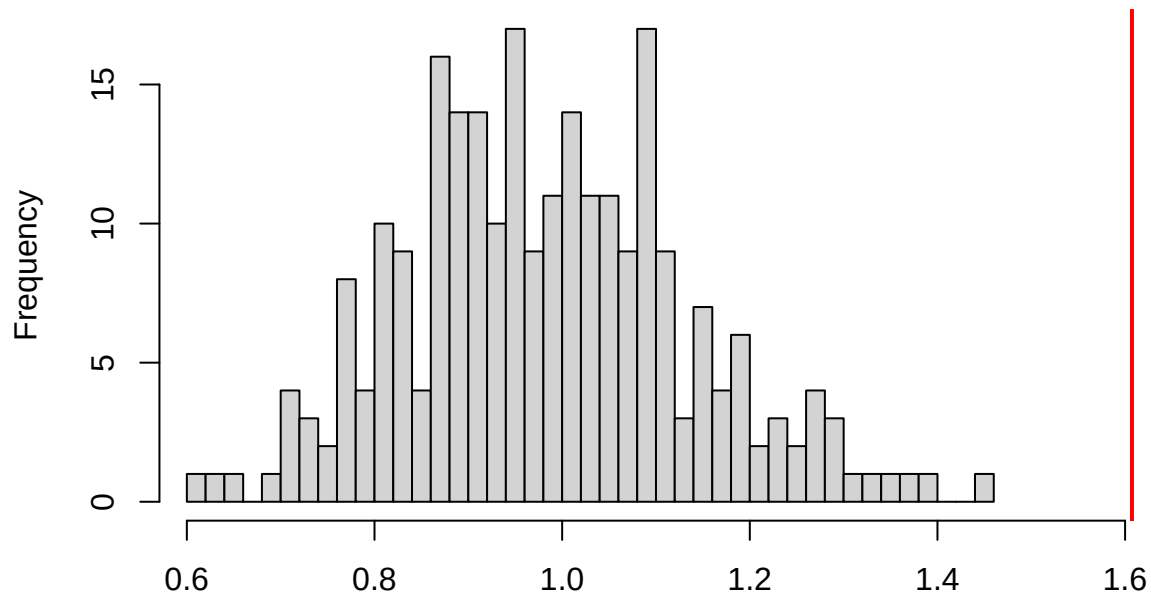
## Natural lacerate production analysis

mod_tub_pois <- glmer(
  n_lacerates ~ temp + (1 | cohort),
  data = lacerates_per_tub,
  family = poisson(link = "log"),
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)

sim_tub <- simulateResiduals(mod_tub_pois)
print(testDispersion(sim_tub))

```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



Simulated values, red line = fitted model. p-value (two.sided) = 0

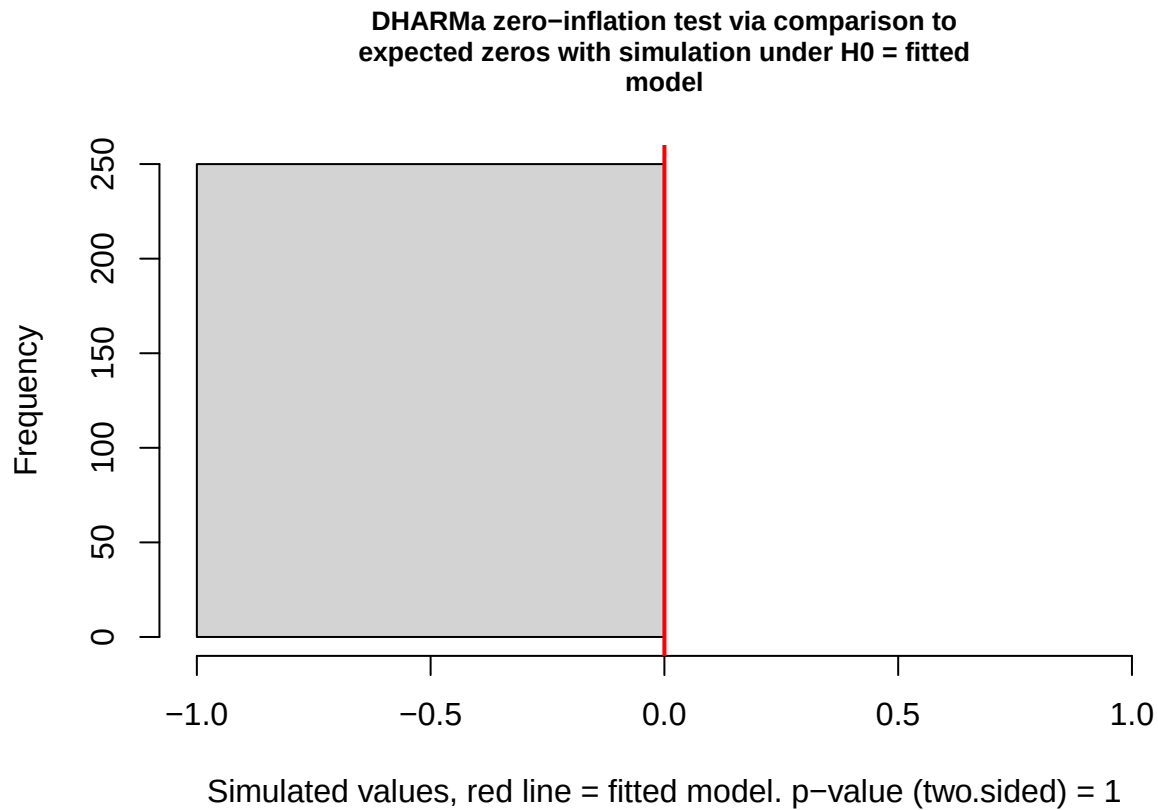
```

##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data:  simulationOutput
## dispersion = 1.6356, p-value < 2.2e-16

```

```
## alternative hypothesis: two.sided
```

```
print(testZeroInflation(sim_tub))
```



```
##  
## DHARMA zero-inflation test via comparison to expected zeros with  
## simulation under H0 = fitted model  
##  
## data: simulationOutput  
## ratioObsSim = NaN, p-value = 1  
## alternative hypothesis: two.sided
```

```
overdisp_tub <- performance::check_overdispersion(mod_tub_pois)  
print(overdisp_tub)
```

```
## # Overdispersion test (using simulated residuals)  
##  
## dispersion ratio = 1.636  
## p-value = < 0.001
```

```
if (overdisp_tub$dispersion_ratio > 1.5) {  
  mod_tub_final <- glmmTMB(  
    n_lacerates ~ temp + (1 | cohort),  
    data = lacerates_per_tub,
```

```

    family = nbinom2()
  )
} else {
  mod_tub_final <- mod_tub_pois
}

print(Anova(mod_tub_final, type = "II"))

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: n_lacerates
##      Chisq Df Pr(>Chisq)
## temp 0.6243  1    0.4295

emm_tub <- emmeans(mod_tub_final, ~ temp, type = "response")
print(emm_tub)

```

```

## temp response SE df asymp.LCL asymp.UCL
## 25      33.7 6.85 Inf      22.6      50.2
## 32      30.2 6.14 Inf      20.2      45.0
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale

```

```

print(pairs(emm_tub))

```

```

## contrast      ratio SE df null z.ratio p.value
## temp25 / temp32 1.12 0.156 Inf  1  0.790 0.4295
##
## Tests are performed on the log scale

```

```

## Final tentacle count

```

```

lacerate_counts <- read_csv(
  file.path(project_dir, "data/natural_lacerate_counts.csv"),
  show_col_types = FALSE
) %>%
  clean_names() %>%
  mutate(
    date = ymd(date),
    cohort = factor(cohort, levels = cohort_levels),
    treatment = factor(treatment, levels = c("H2_25", "H2_32")),
    genotype = factor(genotype, levels = "H2"),
    temp = factor(temp, levels = c("25", "32")),
    lacerate_id = as.character(lacerate_id),
    tentacle_count = as.numeric(tentacle_count)
  )

final_tentacle_dat <- lacerate_counts %>%
  filter(!is.na(tentacle_count)) %>%
  arrange(lacerate_id, cohort, date) %>%

```

```

group_by(lacerate_id, cohort, treatment, genotype, temp) %>%
slice_tail(n = 1) %>%
ungroup()

print(final_tentacle_dat)

```

```

## # A tibble: 1,024 x 17
##   date      session observer lacerate_id  days_since_formation tentacle_count
##   <date>    <chr>    <chr>    <chr>          <dbl>          <dbl>
## 1 2026-04-06 Mon      Grady    P1_H2_25_Tub~      10             18
## 2 2026-04-06 Mon      Grady    P1_H2_25_Tub~      10             15
## 3 2026-04-06 Mon      Grady    P1_H2_25_Tub~      10              7
## 4 2026-04-06 Mon      Grady    P1_H2_25_Tub~      10             10
## 5 2026-04-06 Mon      Grady    P1_H2_25_Tub~       0             10
## 6 2026-04-06 Mon      Grady    P1_H2_25_Tub~       0              7
## 7 2026-04-06 Mon      Grady    P1_H2_25_Tub~       0              4
## 8 2026-04-06 Mon      Grady    P1_H2_25_Tub~       0              0
## 9 2026-03-09 Mon      Grady    P1_H2_25_Tub~      10              6
## 10 2026-04-20 Mon      Grady    P1_H2_25_Tub~      13             14
## # i 1,014 more rows
## # i 11 more variables: alive_status <chr>, notes <chr>, cohort <fct>,
## #   rate2 <chr>, modified_days_since_formation <dbl>, rate <dbl>,
## #   genotype <fct>, temp <fct>, parent_id <chr>, tub_id <chr>, treatment <fct>

```

```

mod_tentacle <- glmmTMB(
  tentacle_count ~ temp * cohort,
  family = nbinom2(),
  data = final_tentacle_dat
)

print(Anova(mod_tentacle, type = "II"))

```

```

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: tentacle_count
##              Chisq Df Pr(>Chisq)
## temp          41.4629  1  1.201e-10 ***
## cohort         3.7109  4  0.4465432
## temp:cohort    21.0936  4  0.0003034 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

emm_tentacle <- emmeans(
  mod_tentacle,
  ~ temp | cohort,
  type = "response"
)

print(emm_tentacle)

```

```

## cohort = W1-2:
## temp response      SE df asymp.LCL asymp.UCL

```

```
## 25      6.06 0.710 Inf      4.82      7.62
## 32      8.86 0.807 Inf      7.41     10.59
##
## cohort = W3-4:
## temp response      SE df asymp.LCL asymp.UCL
## 25      9.90 0.746 Inf      8.54     11.48
## 32      7.04 0.443 Inf      6.23      7.97
##
## cohort = W5-6:
## temp response      SE df asymp.LCL asymp.UCL
## 25     10.67 0.586 Inf      9.58     11.89
## 32      7.50 0.516 Inf      6.55      8.58
##
## cohort = W7-8:
## temp response      SE df asymp.LCL asymp.UCL
## 25     10.32 0.561 Inf      9.28     11.48
## 32      7.39 0.533 Inf      6.42      8.52
##
## cohort = W9-10:
## temp response      SE df asymp.LCL asymp.UCL
## 25      9.66 0.456 Inf      8.81     10.60
## 32      7.29 0.419 Inf      6.52      8.16
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

```
print(pairs(emm_tentacle, adjust = "tukey"))
```

```
## cohort = W1-2:
## contrast      ratio      SE df null z.ratio p.value
## temp25 / temp32 0.684 0.1010 Inf  1  -2.561  0.0104
##
## cohort = W3-4:
## contrast      ratio      SE df null z.ratio p.value
## temp25 / temp32 1.406 0.1380 Inf  1   3.468  0.0005
##
## cohort = W5-6:
## contrast      ratio      SE df null z.ratio p.value
## temp25 / temp32 1.423 0.1250 Inf  1   4.007 <0.0001
##
## cohort = W7-8:
## contrast      ratio      SE df null z.ratio p.value
## temp25 / temp32 1.396 0.1260 Inf  1   3.692  0.0002
##
## cohort = W9-10:
## contrast      ratio      SE df null z.ratio p.value
## temp25 / temp32 1.324 0.0985 Inf  1   3.780  0.0002
##
## Tests are performed on the log scale
```

```
cld_tentacle_df <- as.data.frame(
  multcomp::cld(
    emm_tentacle,
```

```

    Letters = letters,
    adjust = "tukey"
  )
) %>%
mutate(
  treatment = factor(
    paste0("H2_", temp),
    levels = c("H2_25", "H2_32")
  ),
  .group = str_trim(.group)
) %>%
left_join(
  final_tentacle_dat %>%
    group_by(cohort) %>%
    summarise(
      y = max(tentacle_count, na.rm = TRUE) * 1.08,
      .groups = "drop"
    ),
  by = "cohort"
)

## Figure S5

p_lacerates_tub_5 <- ggplot(
  lacerates_per_tub,
  aes(x = treatment, y = n_lacerates, color = treatment)
) +
  geom_standard_boxplot() +
  geom_standard_jitter(height = 0.05) +
  geom_text(
    data = letters_tub,
    aes(x = treatment, y = y_pos, label = .group),
    inherit.aes = FALSE,
    size = ANNOT_SIZE,
    fontface = "bold",
    color = "black"
  ) +
  cohort_facet(ncol = 5) +
  coord_cartesian(ylim = c(0, 80), clip = "off") +
  temp_color_scale(c("H2_25", "H2_32")) +
  temp_x_scale(c("H2_25", "H2_32")) +
  labs(
    x = NULL,
    y = "Number of lacerates per tub",
    color = "Temperature"
  ) +
  my_theme

p_final_tentacles_5 <- ggplot(
  final_tentacle_dat,
  aes(x = treatment, y = tentacle_count, color = treatment)
) +

```

```

geom_standard_boxplot() +
geom_standard_jitter() +
geom_text(
  data = cld_tentacle_df,
  aes(x = treatment, y = y, label = .group),
  inherit.aes = FALSE,
  color = "black",
  size = ANNOT_SIZE,
  fontface = "bold"
) +
cohort_facet(ncol = 5) +
coord_cartesian(ylim = c(0, 27), clip = "off") +
temp_color_scale(c("H2_25", "H2_32")) +
temp_x_scale(c("H2_25", "H2_32")) +
labs(
  x = "Temperature",
  y = "Final tentacle count",
  color = "Temperature"
) +
my_theme +
theme(
  legend.position = "right",
  legend.justification = c(0, 1.4),
  legend.box.just = "top"
)

p_lacerate_combined <-
(p_lacerates_tub_5 + guides(color = "none")) /
p_final_tentacles_5 +
plot_layout(ncol = 1, heights = c(1, 1)) +
plot_annotation(tag_levels = "A")

## Symbiont density weeks 7-8

lacerate_sym_density <- read_csv(
  file.path(project_dir, "data/natural_lacerate_sym_density_W7-8.csv"),
  show_col_types = FALSE
) %>%
clean_names() %>%
mutate(
  label = as.character(label),
  region = as.character(region),
  plate = as.character(plate),
  notes = na_if(str_trim(as.character(notes)), ""),
  calculation = as.character(calculation),
  well = str_trim(as.character(well)),
  type = str_trim(as.character(type)),
  cohort = str_trim(as.character(cohort)),
  symbiotic_state = str_trim(as.character(symbiotic_state)),
  area = parse_number(as.character(area)),
  min_threshold = parse_number(as.character(min_threshold)),
  max_threshold = parse_number(as.character(max_threshold)),

```

```

percent_raw = parse_number(as.character(percent_raw)),
treatment = str_trim(as.character(treatment)),
genotype = str_trim(as.character(genotype)),
temperature = str_trim(as.character(temperature))
) %>%
filter(
  treatment != "#VALUE!",
  genotype != "#VALUE!",
  temperature != "#VALUE!",
  is.na(notes)
) %>%
mutate(
  treatment = factor(treatment, levels = c("H2-25", "H2-32")),
  genotype = factor(genotype, levels = "H2"),
  temperature = factor(temperature, levels = c("25", "32")),
  cohort = factor(cohort, levels = "W7-8")
)

symbiont_area_only <- lacerate_sym_density %>%
  filter(type == "Symbiont Area") %>%
  mutate(
    calculation_num = parse_number(as.character(calculation))
  )

symbiont_area_well_avg <- symbiont_area_only %>%
  group_by(cohort, treatment, genotype, temperature, well) %>%
  summarise(
    mean_calculation = mean(calculation_num, na.rm = TRUE),
    n_images = n(),
    .groups = "drop"
  )

mod_sym_calc <- lm(
  mean_calculation ~ temperature,
  data = symbiont_area_well_avg
)

print(Anova(mod_sym_calc, type = "II"))

## Anova Table (Type II tests)
##
## Response: mean_calculation
##           Sum Sq Df F value    Pr(>F)
## temperature 18339  1 102.34 5.733e-16 ***
## Residuals   14335 80
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

emm_sym_calc <- emmeans(mod_sym_calc, ~ temperature)
print(emm_sym_calc)

## temperature emmean    SE df lower.CL upper.CL

```

```
## 25          54.6 2.07 80      50.5      58.7
## 32          24.7 2.12 80      20.4      28.9
##
## Confidence level used: 0.95
```

```
print(pairs(emm_sym_calc, adjust = "tukey"))
```

```
## contrast          estimate    SE df t.ratio p.value
## temperature25 - temperature32      29.9 2.96 80  10.116 <0.0001
```

```
cld_sym_calc_df <- as.data.frame(
  multcomp::cld(
    emm_sym_calc,
    Letters = letters,
    adjust = "tukey"
  )
) %>%
mutate(
  treatment = factor(
    paste0("H2-", temperature),
    levels = c("H2-25", "H2-32")
  ),
  .group = str_trim(.group)
)

sym_calc_ypos <- symbiont_area_well_avg %>%
  group_by(treatment) %>%
  summarise(
    y = max(mean_calculation, na.rm = TRUE) * 1.08,
    .groups = "drop"
  )

cld_sym_calc_df <- cld_sym_calc_df %>%
  left_join(sym_calc_ypos, by = "treatment")

p_symbiont_calc <- ggplot(
  symbiont_area_well_avg,
  aes(x = treatment, y = mean_calculation, color = treatment)
) +
  geom_standard_boxplot() +
  geom_standard_jitter() +
  geom_text(
    data = cld_sym_calc_df,
    aes(x = treatment, y = y, label = .group),
    inherit.aes = FALSE,
    color = "black",
    size = ANNOT_SIZE,
    fontface = "bold"
  ) +
  cohort_facet() +
  temp_color_scale(c("H2-25", "H2-32")) +
  temp_x_scale(c("H2-25", "H2-32")) +
  labs(
```

```

    x = "Temperature",
    y = "Symbiont density (%)",
    color = "Temperature"
  ) +
  my_theme +
  theme(legend.position = "none")

## Exact p-value figures

## Natural lacerates per tub

pvals_lacerates <- lacerates_per_tub %>%
  group_by(cohort) %>%
  group_modify(~ {
    dat <- .x
    mod <- lm(n_lacerates ~ temp, data = dat)
    em <- emmeans(mod, ~ temp)

    contrast_result <- pairs(
      em,
      adjust = "sidak"
    ) %>%
      as.data.frame()

    tibble(p.value = contrast_result$p.value[1])
  }) %>%
  ungroup() %>%
  left_join(
    lacerates_per_tub %>%
      group_by(cohort) %>%
      summarise(
        max_y = max(n_lacerates, na.rm = TRUE),
        .groups = "drop"
      ),
    by = "cohort"
  ) %>%
  mutate(
    y_bracket = max_y + 5,
    y_text = max_y + 8,
    p_label = format_exact_p(p.value)
  )

p_lacerates_tub_p <- ggplot(
  lacerates_per_tub,
  aes(x = treatment, y = n_lacerates, color = treatment)
) +
  geom_standard_boxplot() +
  geom_standard_jitter(height = 0.05) +
  add_p_bracket(pvals_lacerates, tick_height = 1.5) +
  cohort_facet(ncol = 5) +
  coord_cartesian(ylim = c(0, 85), clip = "off") +
  temp_color_scale(c("H2_25", "H2_32")) +

```

```

temp_x_scale(c("H2_25", "H2_32")) +
labs(
  x = NULL,
  y = "Number of lacerates per tub",
  color = "Temperature"
) +
my_theme

## Final tentacle count

pvals_tentacles <- pairs(
  emm_tentacle,
  adjust = "tukey"
) %>%
as.data.frame() %>%
select(cohort, p.value) %>%
left_join(
  final_tentacle_dat %>%
    group_by(cohort) %>%
    summarise(
      max_y = max(tentacle_count, na.rm = TRUE),
      .groups = "drop"
    ),
  by = "cohort"
) %>%
mutate(
  y_bracket = max_y + 1.5,
  y_text = max_y + 3,
  p_label = format_exact_p(p.value)
)

p_final_tentacles_p <- ggplot(
  final_tentacle_dat,
  aes(x = treatment, y = tentacle_count, color = treatment)
) +
geom_standard_boxplot() +
geom_standard_jitter() +
add_p_bracket(pvals_tentacles, tick_height = 0.7) +
cohort_facet(ncol = 5) +
coord_cartesian(ylim = c(0, 27), clip = "off") +
temp_color_scale(c("H2_25", "H2_32")) +
temp_x_scale(c("H2_25", "H2_32")) +
labs(
  x = "Temperature",
  y = "Final tentacle count",
  color = "Temperature"
) +
my_theme +
theme(
  legend.position = "right",
  legend.justification = c(0, 1.4),
  legend.box.just = "top"
)

```

```

)

p_lacerate_combined_p <-
  (p_lacerates_tub_p + guides(color = "none")) /
  p_final_tentacles_p +
  plot_layout(ncol = 1, heights = c(1, 1)) +
  plot_annotation(tag_levels = "A")

## Symbiont density

pval_symbiont <- pairs(
  emm_sym_calc,
  adjust = "tukey"
) %>%
  as.data.frame() %>%
  slice(1) %>%
  select(p.value)

symbiont_max <- max(
  symbiont_area_well_avg$mean_calculation,
  na.rm = TRUE
)

pval_symbiont <- pval_symbiont %>%
  mutate(
    cohort = factor("W7-8", levels = "W7-8"),
    y_bracket = symbiont_max + 4,
    y_text = symbiont_max + 8,
    p_label = format_exact_p(p.value)
  )

p_symbiont_calc_p <- ggplot(
  symbiont_area_well_avg,
  aes(x = treatment, y = mean_calculation, color = treatment)
) +
  geom_standard_boxplot() +
  geom_standard_jitter() +
  add_p_bracket(pval_symbiont, tick_height = 2) +
  cohort_facet() +
  coord_cartesian(
    ylim = c(0, symbiont_max + 14),
    clip = "off"
  ) +
  temp_color_scale(c("H2-25", "H2-32")) +
  temp_x_scale(c("H2-25", "H2-32")) +
  labs(
    x = "Temperature",
    y = "Symbiont density (%)",
    color = "Temperature"
  ) +
  my_theme +
  theme(legend.position = "none")

```

Representative images

```
img_meta_symbiont <- tribble(
  ~row_lab, ~col_lab, ~file,
  "25°C", "BF", file.path(project_dir, "figs/images/NL_25_Snap499_bf.png"),
  "25°C", "FL", file.path(project_dir, "figs/images/NL_25_Snap498_fl.png"),
  "32°C", "BF", file.path(project_dir, "figs/images/NL_32_Snap645_bf.png"),
  "32°C", "FL", file.path(project_dir, "figs/images/NL_32_Snap644_fl.png")
)

make_symbiont_image_panel <- function(img_meta) {

  image_size <- 1.28
  col_gap <- 1.31
  row_gap <- 1.38

  img_meta <- img_meta %>%
    mutate(
      row_lab = factor(row_lab, levels = c("32°C", "25°C")),
      col_lab = factor(col_lab, levels = c("BF", "FL")),
      row_num = as.numeric(row_lab),
      col_num = as.numeric(col_lab),
      x_center = col_num * col_gap + 0.34,
      y_center = row_num * row_gap
    )

  p <- ggplot() +
    xlim(0.52, 3.52) +
    ylim(0.28, 3.55) +
    theme_void(base_size = BASE_SIZE, base_family = "sans")

  for (i in seq_len(nrow(img_meta))) {
    if (!file.exists(img_meta$file[i])) {
      warning("Image not found: ", img_meta$file[i])
      next
    }

    img <- image_read(img_meta$file[i])
    grob <- rasterGrob(as.raster(img), interpolate = TRUE)

    p <- p +
      annotation_custom(
        grob,
        xmin = img_meta$x_center[i] - image_size / 2,
        xmax = img_meta$x_center[i] + image_size / 2,
        ymin = img_meta$y_center[i] - image_size / 2,
        ymax = img_meta$y_center[i] + image_size / 2
      )
  }

  row_df <- img_meta %>%
    distinct(row_lab, y_center) %>%
```

```

    mutate(
      lab = paste0("Sym\n", as.character(row_lab))
    )

col_df <- img_meta %>%
  distinct(col_lab, x_center) %>%
  mutate(
    lab = recode(
      as.character(col_lab),
      "BF" = "Brightfield",
      "FL" = "Fluorescent"
    )
  )

p +
  geom_text(
    data = row_df,
    aes(x = 0.93, y = y_center, label = lab),
    hjust = 1,
    size = ANNOT_SIZE,
    fontface = "bold",
    family = "sans"
  ) +
  geom_text(
    data = col_df,
    aes(x = x_center, y = 0.30, label = lab),
    vjust = 1,
    size = ANNOT_SIZE,
    fontface = "bold",
    family = "sans"
  )
}

symbiont_image_panel <- make_symbiont_image_panel(img_meta_symbiont)

## Figure S6

## Letter version

p_symbiont_calc_with_images <-
  wrap_plots(
    p_symbiont_calc,
    symbiont_image_panel,
    ncol = 2,
    widths = c(1.35, 1.00)
  ) +
  plot_annotation(
    tag_levels = "A"
  ) &

theme(
  plot.tag = element_text(

```

```

    family = "sans",
    size = 18,
    face = "bold",
    color = "black"
  ),

  plot.tag.position = c(0.025, 0.985)
)

## Exact p-value version

p_symbiont_calc_p_with_images <-
  wrap_plots(
    p_symbiont_calc_p,
    symbiont_image_panel,
    ncol = 2,
    widths = c(1.35, 1.00)
  ) +
  plot_annotation(
    tag_levels = "A"
  ) &

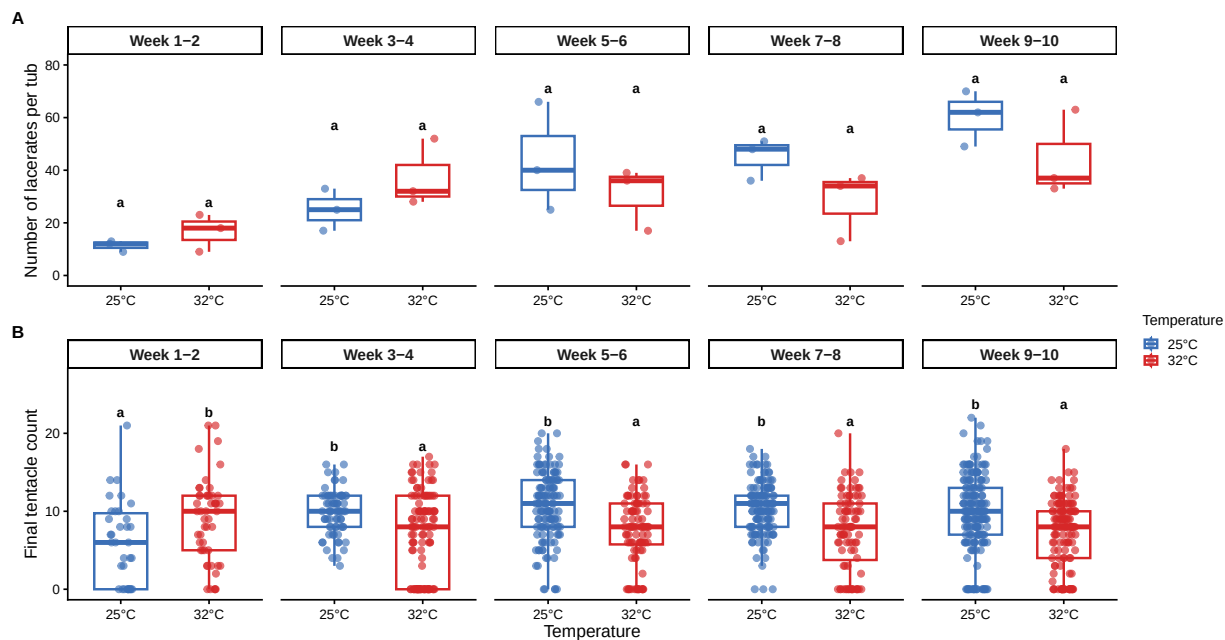
  theme(
    plot.tag = element_text(
      family = "sans",
      size = 18,
      face = "bold",
      color = "black"
    ),

    plot.tag.position = c(0.025, 0.985)
  )

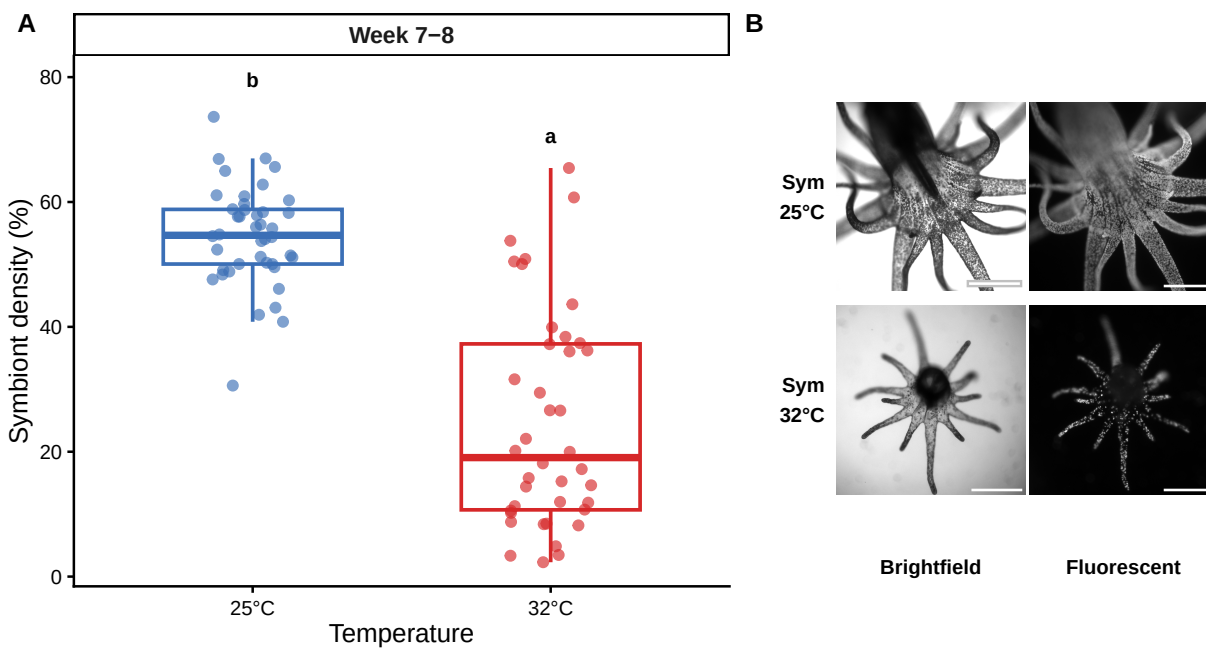
```

Figures

```
print(p_lacerate_combined)
```



```
print(p_symbiont_calc_with_images)
```



```
## Save figures
```

```
ggsave(
  filename = "FigS5.png",
  plot = p_lacerate_combined,
  path = "~/Documents/GitHub/heating_lacerates_final/natural/figs",
```

```

device = "png",
width = 18,
height = 9.5,
units = "in",
dpi = 600,
bg = "white"
)

ggsave(
  filename = "FigS6.png",
  plot = p_symbiont_calc_with_images,
  path = "~/Documents/GitHub/heating_lacerates_final/natural/figs",
  device = "png",
  width = 12,
  height = 6.5,
  units = "in",
  dpi = 600,
  bg = "white"
)

## Supplemental statistics tables

## Table S12: Natural lacerate production, overall temperature effect

s5a_anova <- car::Anova(mod_tub_final, type = "II") %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  as_tibble()

print(s5a_anova)

## # A tibble: 1 x 4
##   Effect Chisq    Df 'Pr(>Chisq)'
##   <chr>   <dbl> <dbl>         <dbl>
## 1 temp    0.624     1         0.429

## Table S13: Natural lacerate production, overall temperature comparison

s5a_pairwise <- pairs(
  emmeans(mod_tub_final, ~ temp, type = "response")
) %>%
  as.data.frame() %>%
  as_tibble()

print(s5a_pairwise)

## # A tibble: 1 x 7
##   contrast      ratio    SE    df null z.ratio p.value
##   <fct>         <dbl> <dbl> <dbl> <dbl>   <dbl>   <dbl>
## 1 temp25 / temp32  1.12 0.156  Inf     1    0.790    0.429

```

```
## Table S14: Natural lacerate production by 2-week interval
```

```
s5a_by_cohort <- lacerates_per_tub %>%
  group_by(cohort) %>%
  group_modify(~ {

    mod <- lm(n_lacerates ~ temp, data = .x)

    pairs(
      emmeans(mod, ~ temp),
      adjust = "sidak"
    ) %>%
      as.data.frame()

  }) %>%
  ungroup() %>%
  select(
    cohort,
    contrast,
    estimate,
    SE,
    df,
    t.ratio,
    p.value
  )

print(s5a_by_cohort)
```

```
## # A tibble: 5 x 7
##   cohort contrast      estimate    SE    df t.ratio p.value
##   <fct> <chr>          <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 W1-2   temp25 - temp32   -5.33  4.27    4  -1.25  0.280
## 2 W3-4   temp25 - temp32  -12.3  8.74    4  -1.41  0.231
## 3 W5-6   temp25 - temp32    13   13.8    4   0.941  0.400
## 4 W7-8   temp25 - temp32    17    8.83    4   1.92  0.127
## 5 W9-10  temp25 - temp32    16.0  11.2    4   1.43  0.227
```

```
## Table S15: Final tentacle count, overall model
```

```
s5b_anova <- car::Anova(mod_tentacle, type = "II") %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  as_tibble()

print(s5b_anova)
```

```
## # A tibble: 3 x 4
##   Effect      Chisq    Df 'Pr(>Chisq)'
##   <chr>      <dbl> <dbl>      <dbl>
## 1 temp      41.5     1  1.20e-10
## 2 cohort     3.71     4  4.47e- 1
## 3 temp:cohort 21.1     4  3.03e- 4
```

```
## Table S16: Final tentacle count by 2-week interval
```

```
s5b_pairwise <- pairs(
  emm_tentacle,
  adjust = "tukey"
) %>%
  as.data.frame() %>%
  as_tibble() %>%
  select(
    cohort,
    contrast,
    ratio,
    SE,
    df,
    z.ratio,
    p.value
  )

print(s5b_pairwise)
```

```
## # A tibble: 5 x 7
##   cohort contrast      ratio      SE    df z.ratio  p.value
##   <fct>   <fct>      <dbl>   <dbl> <dbl>   <dbl>    <dbl>
## 1 W1-2   temp25 / temp32 0.684 0.101   Inf   -2.56 0.0104
## 2 W3-4   temp25 / temp32 1.41 0.138   Inf    3.47 0.000524
## 3 W5-6   temp25 / temp32 1.42 0.125   Inf    4.01 0.0000615
## 4 W7-8   temp25 / temp32 1.40 0.126   Inf    3.69 0.000222
## 5 W9-10  temp25 / temp32 1.32 0.0985   Inf    3.78 0.000157
```

```
## Table S17: Symbiont density, overall temperature effect
```

```
s6_anova <- car::Anova(mod_sym_calc, type = "II") %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  as_tibble()

print(s6_anova)
```

```
## # A tibble: 2 x 5
##   Effect      'Sum Sq'    Df 'F value'  'Pr(>F)'
##   <chr>      <dbl> <dbl>    <dbl>    <dbl>
## 1 temperature 18339.    1    102. 5.73e-16
## 2 Residuals 14335.   80     NA    NA
```

```
## Table S18: Symbiont density, temperature comparison
```

```
s6_pairwise <- pairs(
  emm_sym_calc,
  adjust = "tukey"
) %>%
  as.data.frame() %>%
  as_tibble() %>%
```

```

select(
  contrast,
  estimate,
  SE,
  df,
  t.ratio,
  p.value
)

print(s6_pairwise)

```

```

## # A tibble: 1 x 6
##   contrast          estimate    SE    df t.ratio p.value
##   <chr>          <dbl> <dbl> <dbl>   <dbl>   <dbl>
## 1 temperature25 - temperature32    29.9  2.96    80    10.1 5.73e-16

```